

Relationship between urine bisphenol-A level and declining male sexual function.

The adverse effect of bisphenol-A (BPA) on the male reproductive system observed in animal studies has not been well examined in human populations. BPA is potentially a serious public health problem because of its widely detected presence in the human body. This study was conducted among 427 male workers in regions where high levels of BPA exposure existed. All participants provided urine samples, which were tested for BPA concentration using high-performance liquid chromatography. Male sexual dysfunction was ascertained using standard male sexual function inventories. Male sexual dysfunction was measured in 4 domains using 7 indices. After controlling for potential confounders using linear regression, increasing urine BPA level was associated with worsening male sexual function on a continuous scale. All 7 indices demonstrated this negative linear correlation. Increasing urine BPA level was associated with decreased sexual desire ($P < .001$), more difficulty having an erection ($P < .001$), lower ejaculation strength ($P < .001$), and lower level of overall satisfaction with sex life ($P < .01$). A similar negative correlation was also observed among participants exposed to BPA from only environmental sources (no occupational exposure to BPA), although the estimates in this group were less stable because of a smaller sample size. Our results reveal a correlation between a biological measure of urine BPA level and declining male sexual function. This finding may enhance the understanding of the BPA effect in human populations, and may have important public health implications given the widespread human exposure to BPA.